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SUBJECT-Chemistry

Class-IX

Study of Carbon Monoxide and Carbon Dioxide

INSTRUCTIONS TO THE STUDENTS- **Go through the lesson in 3 days and on the 4th day try to solve the questions given in the self evaluation portion.**

INDEX

Carbon Monoxide

i. Introduction
ii. Occurrence
iii. Preparation
iv. Properties(Physical and Chemical)
v. Tests
vi. Uses
vii. Fire extinguishers
viii. Comparative study between CO and CO_2
ix. Quicklime, slaked lime and limewater

For better understanding of the topic you may refer to the following videos

Chemistry of carbon compounds https://youtu.be/Sh2CXh6xbgQ (courtesy DIGICLASS A)

Preparation of CO_2 in laboratory https://youtu.be/Du8Exmj9sW4 (courtesy Science projects)

CO_2 preparation & properties https://www.youtube.com/watch?v=ZlWE1MIr1RU (courtesy Kimika Malta)

Preparation of CO_2 https://www.youtube.com/watch?v=fC5YQVlS7TI (courtesy DeltaStep)

Physical properties of carbon dioxide https://youtu.be/l6cCloagAuQ (courtesy Deltastep)

Chemical properties of carbon dioxide https://youtu.be/0sQCSB1j3cw (courtesy Deltastep)

Burning of Mg in CO_2 https://youtu.be/2oQ_9nFe9HU (courtesy Royal Society for Chemistry)

Test for CO_2 https://youtu.be/0J4HpjwKwsg (courtesy Ong SK 1984)

Soda acid type fire extinguisher https://youtu.be/kfe_72ysJYE (courtesy CBSE)

Foam type portable extinguisher https://youtu.be/RDLEQPNspfg (courtesy marine engineering knowledge)

Foam type fire extinguisher https://youtu.be/VYqBDQHOdT0 (courtesy marine engineering knowledge)

Carbon cycle https://youtu.be/NzxckRZ58JQ (courtesy 7activestudio)

Introduction- In 1630 Van Helmont discovered the gas when charcoal was burnt in air.It occurs in the atmosphere freely t the extent of 0.03% by volume and hence called "fixed air."

In 1873 Lavoisier formulated its basic structure and found it to be an oxide of carbon and initially named it "acid carbonique" and later carbon dioxide.

Occurrence-In free state carbon dioxide constitutes about 0.03% of atmosphere by volume.The percentage of carbon dioxide is higher in volcanic regions and coal mines.

CO_2 is partly soluble in water and partly remains dissolved in natural water.

In combined state Carbon dioxide is found in metallic carbonates and bicarbonate in earth's crust especially in limestone regions containing calcium carbonate($CaCO_3$)

Mineral carbonate in the form of calcium carbonate is also found in sea shells or in coral reefs with layers of dolomite ($MgCO_3.CaCO_3$) embedded in the earth's crust

Preparation of CO_2- General Methods

From carbon compounds

When carbon compounds like charcoal,wood or organic substances are burnt in air or oxygen CO_2 is formed.

Coke(charcoal) $C + O_2 \longrightarrow CO_2 + \triangle$

Hydrocarbons (Acetylene) $2C_2H_2 + 5O_2 \longrightarrow 4CO_2 + 2H_2O + \triangle$

(Benzene) $2C_6H_6 + 15O_2 \longrightarrow 12CO_2 + H_2O + \triangle$

Organic Compounds(Alcohols) $C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$

Thermal Decomposition of metallic carbonates or bicarbonates

Metallic carbonate / bicarbonate								
Calcium carbonate	$CaCO_3$	$\xrightarrow{\Delta}$	CaO	+	CO_2		
Zinc carbonate	$ZnCO_3$	$\xrightarrow{\Delta}$	ZnO	+	CO_2		
Lead carbonate	$PbCO_3$	$\xrightarrow{\Delta}$	PbO	+	CO_2		
Copper carbonate	$CuCO_3$	$\xrightarrow{\Delta}$	CuO	+	CO_2		
Potassium bicarbonate	$2KHCO_3$	$\xrightarrow{\Delta}$	K_2CO_3	+	CO_2	+	H_2O
Sodium bicarbonate	$2NaHCO_3$	$\xrightarrow{\Delta}$	Na_2CO_3	+	CO_2	+	H_2O
Calcium bicarbonate	$Ca(HCO_3)_2$	$\xrightarrow{\Delta}$	$CaCO_3$	+	CO_2	+	H_2O
Magnesium bicarbonate	$Mg(HCO_3)_2$	$\xrightarrow{\Delta}$	$MgCO_3$	+	CO_2	+	H_2O

Action of dilute mineral acids on carbonates and bicarbonates

		Acid [dil.]		Salt		Carbon dioxide		
Metallic bicarbonate								
$NaHCO_3$	+	HCl	→	NaCl	+	CO_2	+	H_2O
$Ca(HCO_3)_2$	+	$2HNO_3$	→	$Ca(NO_3)_2$	+	$2CO_2$	+	$2H_2O$
Metallic carbonate								
Na_2CO_3	+	H_2SO_4	→	Na_2SO_4	+	CO_2	+	H_2O
$CaCO_3$	+	2HCl	→	$CaCl_2$	+	CO_2	+	H_2O

Laboratory preparation

Reactants- Calcium carbonate and dilute Hydrochloric acid

Reaction- $CaCO_3 + 2HCl \longrightarrow CaCl_2 + CO_2 + H_2O$

Purification- During preparation the gas is purified by passage through $KHCO_3$ solution which absorbs residual HCl vapours and through conc. H_2SO_4 which absorbs moisture.

Collection- CO2 is then collected by upward displacement of air

[Note:- Dil H_2SO4 is not used in place of dil HCl since a coating of insoluble CaSO4 is formed on the marble chips and the reaction comes to stop.]

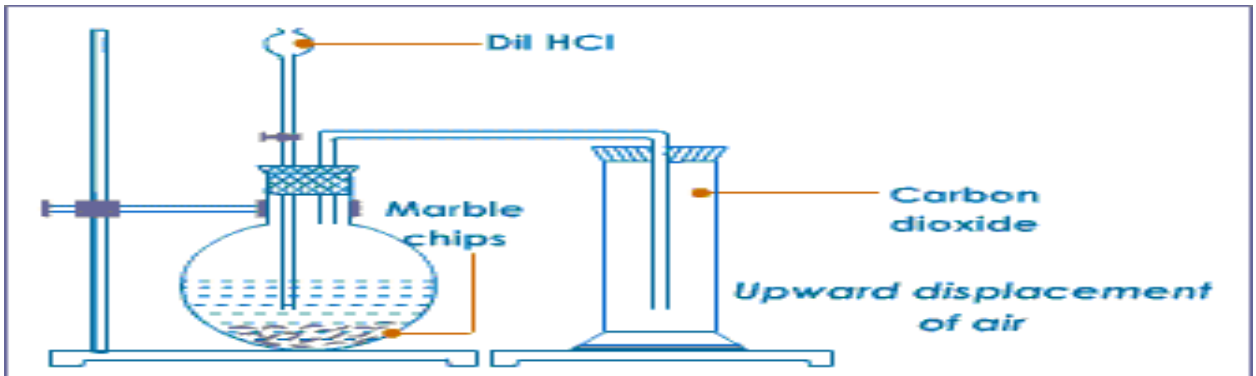

Fig. 11.6 Laboratory Preparation of carbon dioxide

Physical Properties

1. CO_2 is colourless,odourless,non poisonous gas with a slightly sour taste
2. It is about 1.5 times heavier than air and fairly soluble in water
3. It is easily liquefied at about 70 atmospheric pressure and atordinary temperatures.it solidifies on cooling to form a white solid commonly called dry ice
4. Carbon dioxide is non combustible and a non supporter of combustion.It is slightly acidic in nature and does not undergo thermal decompostion except at high temperatures.

Chemical Properties

Reactionsof carbon dioxide

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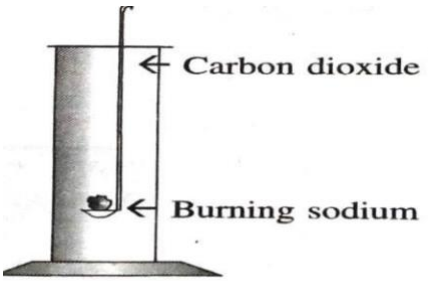
Reactants	Reaction	Equation	Product/s
1. **Water**	Dissolves in water *under pressure.*	$CO_2 + H_2O \rightarrow H_2CO_3$	Carbonic acid [H_2CO_3] [CO_2 - anhydride of carbonic acid]
2. **Ammonia**	Reacts with NH_3 gas at about 200°C.	$2NH_3 + CO_2 \rightarrow (NH_2)_2CO + H_2O$	Urea - $(NH_2)_2$ CO [nitrogenous fertilizer]
3. **Carbon**	Passage over C [coke] reduces it to CO.	$CO_2 + C \rightarrow 2CO$	Carbon monoxide
4. **Metals**			
• **Potassium**	Burning K, Na & Ca burn in carbon dioxide. [Oxides formed react to form carbonates]	$4K + CO_2 \rightarrow 2K_2O + C$	Potassium oxide & carbon
• **Sodium**		$4Na + CO_2 \rightarrow 2Na_2O + C$	Sodium oxide & carbon
• **Calcium**		$2Ca + CO_2 \rightarrow 2CaO + C$ [$CaO + CO_2 \rightarrow CaCO_3$]	Calcium oxide & carbon
• **Magnesium**	Burning Mg, and Al burn in carbon dioxide. [Oxides formed do not react to form carbonates.]	$2Mg + CO_2 \rightarrow 2MgO + C$	Magnesium oxide & carbon
• **Aluminium**		$4Al + 3CO_2 \rightarrow 2Al_2O_3 + 3C$	Aluminium oxide & carbon

Carbon dioxide does not support combustion even then metals like burning sodium or magnesium burns in carbon dioxide as these metals have high heat of combustion and



decomposes carbon dioxide in carbon and oxygen and oxygen supports combustion

Burning of Mg and Na in CO2

- **Carbon dioxide is an acidic oxide so it reacts with water to form a weakacid- carbonic acid**

  $CO_2 + H_2O \longrightarrow H_2CO_3$ **(weak acid)**

- **Acidic oxide reacts with alkali to form salt and water**

Alkali solutions	*Carbon dioxide*	*Carbonate [salt]*	*Water*		
a] Limited CO_2 passed					
• **Potassium hydroxide** $2KOH$ +	CO_2 →	K_2CO_3 + [soluble]	H_2O		
• **Sodium hydroxide** $2NaOH$ +	CO_2 →	Na_2CO_3 + [soluble]	H_2O		
• **Calcium hydroxide** [lime water solution] $Ca(OH)_2$ +	CO_2 →	$CaCO_3\downarrow$ + [insoluble]	H_2O		
b] Excess carbon dioxide passed through the above solutions		K_2CO_3 +	H_2O +	CO_2 →	$2KHCO_3$
		Na_2CO_3 +	H_2O +	CO_2 →	$2NaHCO_3$
		$CaCO_3$ +	H_2O +	CO_2 →	$Ca(HCO_3)_2$

Q.What happens when carbon dioxide is passed through lime water solution?

Carbon dioxide on passage through lime water solution turns it milky but on passage of excess carbon dioxide the milkiness disappears.

1. When limited amount of carbon dioxide is passed through lime water ,it turns milky due to formation of insoluble calcium carbonate($CaCO_3$)

2. If excess of carbon dioxide is passed through milky suspension the milkiness disappears due to formation of water soluble calcium bicarbonate $Ca(HCO_3)_2$.

Q.Why is Potassium hydroxide is preferred to sodium hydroxide for absorption of carbon dioxide?

When excess carbon dioxide is passed through potassium or sodium hydroxide solutions are respective bicarbonates are formed of which sodium bicarbonate is sparingly soluble and may get precipitated, while potassium bicarbonate is completely soluble and hence leaves no precipitate. Therefore potassium hydroxide is preferred to sodium hydroxide for absorption of carbon dioxide.

Tests for CO_2

Test	Observation
1.On passage through lime Water	Lime water turns milky.On passing excess Carbon dioxide through it the milkliness disappears. $Ca(OH)_2+CO_2 \longrightarrow CaCO_3 \downarrow + H_2O$ $CaCO_3+H_2O+CO_2$ (excess) $\longrightarrow Ca(HCO_3)_2$
2.On burning magnesium ribbon In a jar of carbon dioxide	Burns in the gas depositing black particles of carbon on the walls of the jar. $CO_2+ Mg \longrightarrow MgO+C$

[Note: The test with lime water is also observed with SO_2 but SO_2 turns potassium permanganate paper from purple to colourless but CO_2 does not.]

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Uses of CO_2

Industrial uses

1. Manufacture of Urea- **By reaction between gaseous ammonia and carbon dioxide at 200° C and 100 to 20 atm pressure urea is formed.It is animportant nitrogenous fertilizer used in agriculture**

$$2NH_3 + CO_2 \longrightarrow NH_2.CO.NH_2 \text{ (urea)} + H_2O$$

2. Manufacture of baked food items- **Baking powder is prepared by reaction between Baking soda($NaHCO_3$),Potassium hydrogen tartarate andstarch.Baking powder when mixed with dough liberates which makes the dough light and porous.**

3. Usage as dry ice- **Liquid CO_2 on sudden evaporation results in freezing and formation of solid carbon dioxide –dry ice.It is used as refrigerant for preservation of foods.**

4. Aerated drinks- **CO_2gives a fizz to the effervescent drink since on opening the bottle CO_2 escapes out due to the fall in pressure**

Soda-**prepared by passage of CO_2 into water under pressure**

Soft drinks-**also prepared by passage of CO_2 through water under pressure and contains additional sugar, flavours and colours**

5. Fire extinguisher-**CO_2in a saturated solution ejects out and extinguishes fire.$NaHCO_3$ and conc H_2SO_4 constitutes the soda acid fire extinguisher.**

Medical uses

1. CO_2 in from of carbogen stimulates respiration and enhances breathing capacity and is used in cases of gas poisoning and suffocation

2. Mixtures of citric or tartaric acid with sodium bicarbonate is known ashealth salt which on addition of water releases carbon dioxide.

It is used in effervescent drinks and relieves gas pressure in stomach

General uses

1. CO_2 is used in hardening of mortar as atmospheric CO_2 converts the slaked lime into hardened calcium carbonate which binds the brick and sandtogether.

2. In the process of oxygen regeneration in the atmosphere byphotosynthesis atmospheric CO_2 is absorbed by the plants.

Fire Extinguishers

- Carbon dioxide is a non combustible gas and also is a non supporter of combustion.
- Being heavier than air it sinks below a layer of air thus by blanketing the fire and cutting the supply of air which sustains the fire.

Thus CO2 finds usage in fire extinguishers

The two important fire extinguishers are – 1.Soda acid fire extinguisher

2.Foam based fire extinguisher.

The working principal of fire extinguishers is based on the following conditions:

1. Cooling off the combustible material below its ignition temp

2. Cutting off air supply.

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SODA ACID FIRE EXTINGUISHER

Soda acid fire extinguishers comprises of sodium bicarbonate and conc sulphuric acid.It is most efficient house hold fire extinguisher.

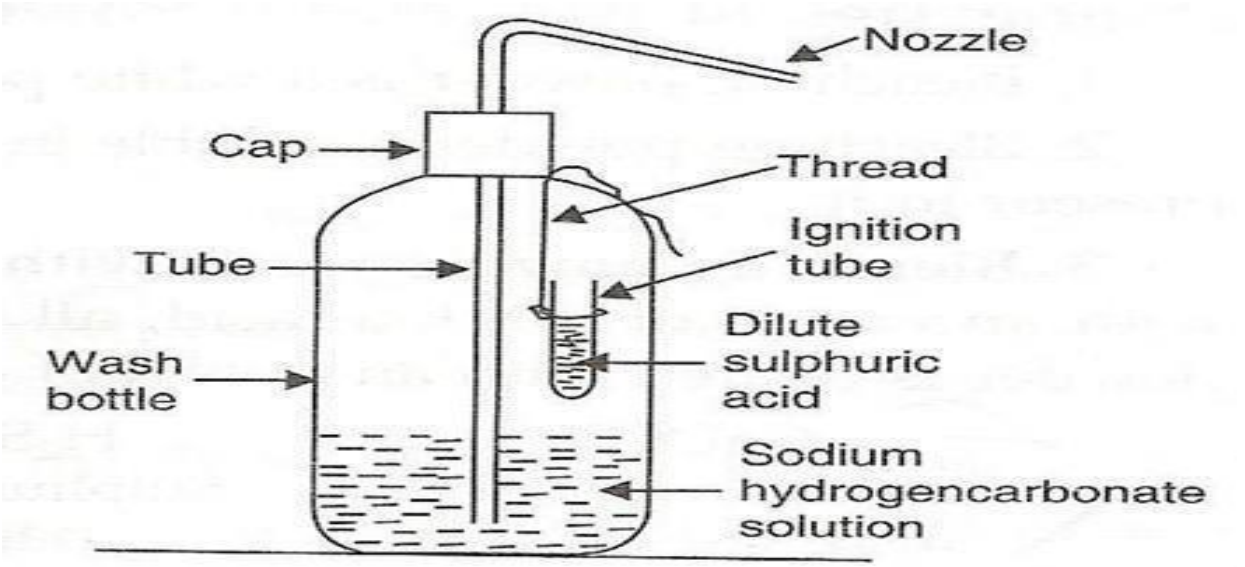
Structure-It consists of a strong iron vessel filled with sodium bicarbonate solution and the vessel has a side discharge nozzle.

Working-In case of a fire the plunger is hit on the floor which breakes the bottle

Conc sulphuric acid mixes with $NaHCO_3$ solution liberating CO_2 gas under pressure.The solution along with CO_2 is injected out with force through the nozzle in saturated form.

Reaction involved-

$$2NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2$$



Making a soda-acid fire extinguisher.

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Foam based FIRE EXTINGUISHER

Foam based fire extinguishers comprises of Sodium bicarbonate ,saponin(for generating foam) and aluminium sulphate solution $Al_2(SO_4)_3$.

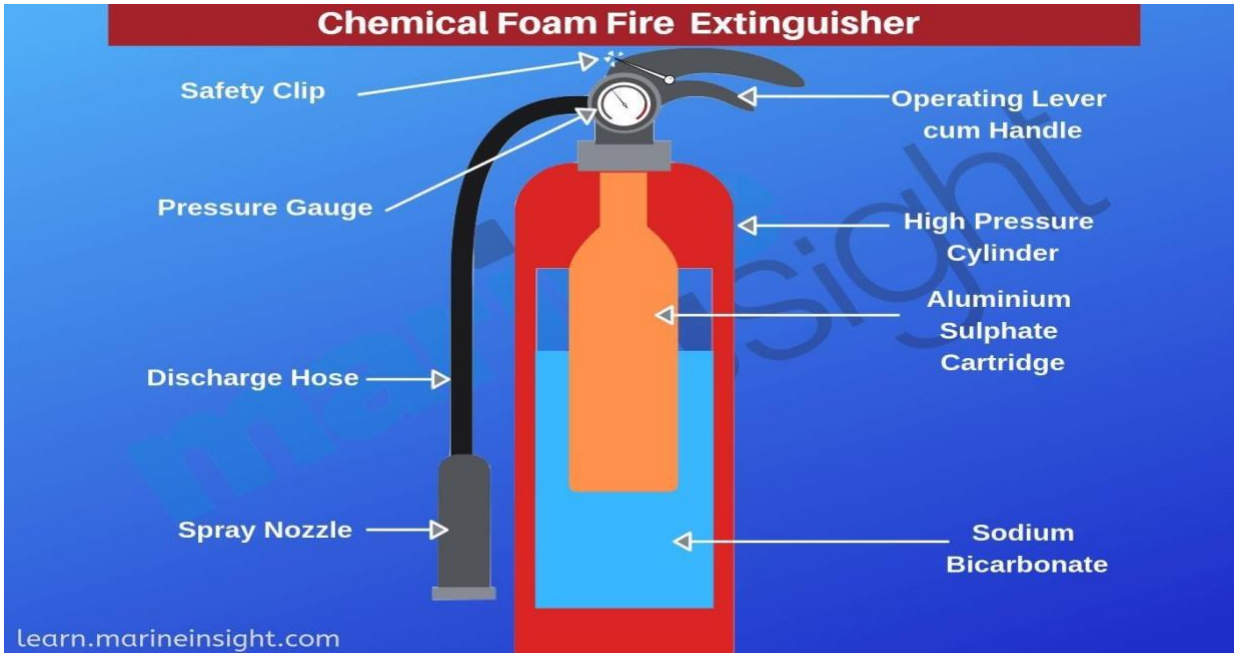
Structure-It consists of a strong iron vessel filled with sodium bicarbonate and saponin and a plunger at the lower end is attached to which a glass bottle containing aluminium sulphate solution is connected.

Working-In case of a fire the plunger hit on the floor and breaks the bottle

Aluminium sulphate mixes with sodium bicarbonate solution liberting CO_2 gas and aluminium hydroxide which together are ejected out under pressure through the nozzle in form of foam.

Reaction involved-

$$6NaHCO_3 + Al_2(SO_4)_3 \rightarrow 3Na_2SO_4 + 2Al(OH)_3 + 6CO_2$$



Q. A foam type fire extinguisher can be used for fighting an oil fire but a soda acid type can not.Why?

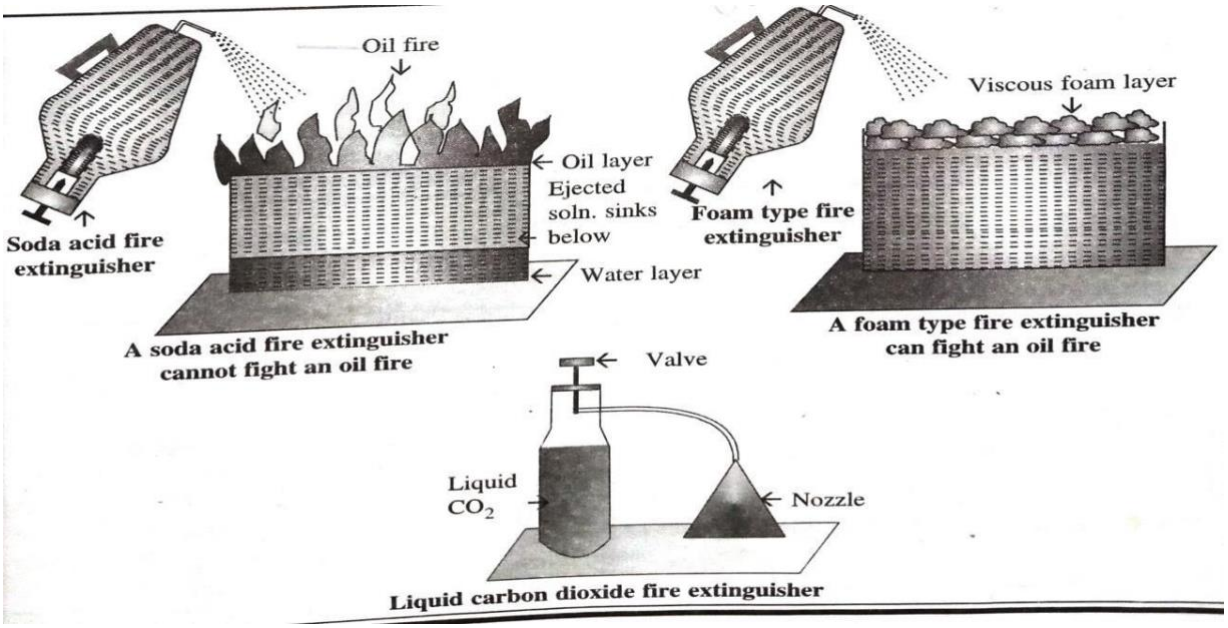
In case of soda acid fire extinguisher the solution ejected out sinks down below the oil fire and hence the air supply is not cut off and fire is not extinguished. Whereas in foam type fire extinguisher the viscous foam ejected out covers the burning oil fire and extinguishes fire by cutting off the air supply.

Q.Both soda type and acid type fire extinguishers can not be used for fighting electrical fires.Why?

In both the above type of fire extinguishers the resultant solution contains sodium sulphate and sodium bicarbonate which being electrolyte themselves may conduct electricity resulting in an electric shock or may enhance short circuiting which may cause another fire.

Q.Liquid carbon dioxide fire extinguisher may be used for both electrical and oil fires.Why?

In liquid carbon dioxide fire extinguisher liquid carbon dioxide under pressure is stored in a steel cylinder on opening valves which cause fall in pressure which causes the liquid carbon dioxide to solidify and be liberated in the form of a white snow which can be used for any of the two types of fire.



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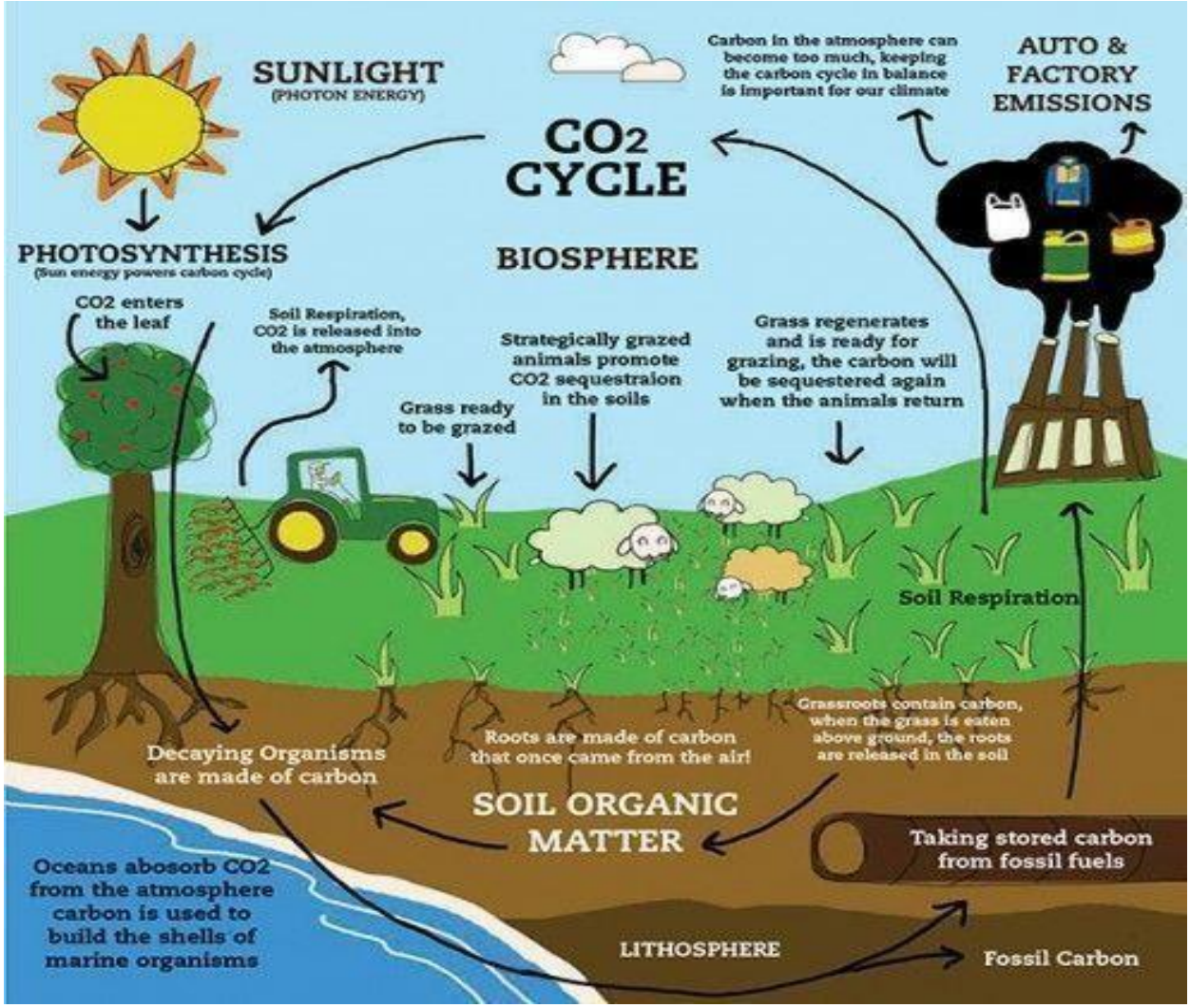
CArbon Cycle

The various process by which the percentage of carbon dioxide in the atmosphere remains constant are collectively represented as the carbon cycle.

Thus large amount of CO2 are added in air and same amount of CO2 is removed thereby maintaining the percentage of carbon dioxide in air at 0.03%



The processes involved in addition and removal of CO_2 in atmosphere

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A Comparative study of CO and CO2

I. CONVERSIONS OF CARBON DIOXIDE AND CARBON MONOXIDE

Conversion	*Method*	*Equation*
1. **Carbon dioxide to carbon monoxide**	By passage of carbon dioxide over heated red hot coke	$CO_2 + C \xrightarrow{[\text{red hot}]} 2CO$
2. **Carbon monoxide to carbon dioxide**	By burning of CO in oxygen	$2CO + O_2 \rightarrow 2CO_2$
	By passage of CO over heated metallic oxides [reduction reaction]	$CO + CuO \xrightarrow{[\text{heated}]} Cu + CO_2$

II. SEPARATION OF CARBON DIOXIDE & CARBON MONOXIDE FROM THEIR MIXTURE

Method - Passage of mixture through	*Equation*
Concentrated potassium hydroxide solution :	
• CO_2 dissolves forming potassium bicarbonate.	$KOH + CO_2 \rightarrow KHCO_3$
• CO remains undissolved and is collected.	
• Recovery of CO_2 from $KHCO_3$ by addition of acid.	$2KHCO_3 + H_2SO_4 \rightarrow K_2SO_4 + 2H_2O + 2CO_2$
Water under pressure	
• CO_2 dissolves forming carbonic acid.	$CO_2 + H_2O \rightarrow H_2CO_3$
• CO remains undissolved and is collected.	
• Recovery of CO_2 from H_2CO_3 by warming	$H_2CO_3 \rightarrow H_2O + CO_2$

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Comparison between CO and CO_2 on basis of their properties

Property	CO	CO2
Vapour Density	V.D=14 Slightly lighter than air	V.D=22 1.5 Times heavier than air
Solubility	Very slightly soluble in water	Fairly soluble in water
Liquefaction	Difficult to liquefy	Easy to liquefy
Nature of Oxide	neutral oxide poisonous in nature	Acidic oxide non poisonous in nature
Litmus test	No effect on litmus	Blue litmus turns pink or Scarlet
Combustibility	Combustible but non supporter of combustion	Noncombustible and non supporter of combustion
Lime water	Not absorbed in lime water	Absorbed in lime water and turns it milky
Oxidizing and reducing nature	Reduces heated oxides of metals	Oxidizes heated metals to oxides

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Quicklime, Slaked Lime and Lime water

Quicklime(CaO)

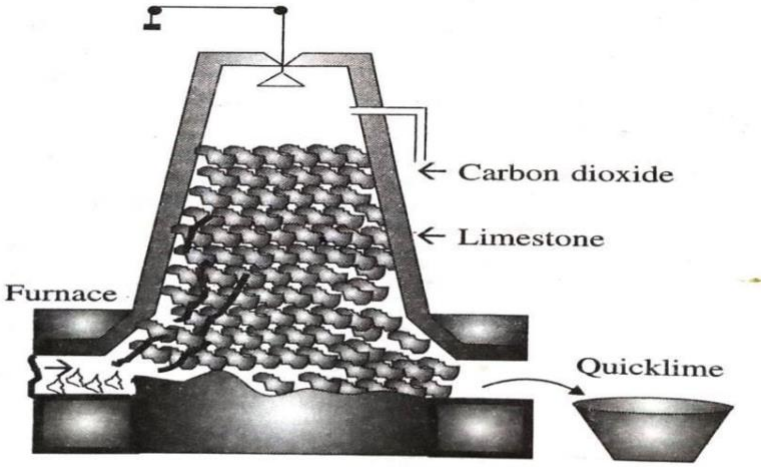
Preparation

Method: By thermal decomposition of limestone $CaCO_3$ in a klin

Equation $CaCO_3$(limestone) $\xrightleftharpoons{\triangle}$ CaO (quicklime)+CO_2

Procedure

- The charge i.e pulverized(crushed) limestone is added from the top of the klin.
- The limestone is heated by the fuel –producer gas which decomposes the limestone into quicklime and carbon dioxide
- The draft of air sweeps the carbon dioxide formed ,out from the top of the klin
- **Product** The quicklime moves out from the lower vent and is collected



Manufacture of quicklime

Properties

Nature: Pure quicklime is a white amorphous powder

Stability: Highly stable to heat and does not decompose.It melts around 1650°C and emits a bright white light called "lime light" when heated in a Oxyhydrogen flame.

Solubility: On addition of water the quicklime swells,crumbles,the water is turned to steam since the reaction is exothermic .The process is called "slaking of lime"

$$CaO + H_2O \longrightarrow Ca(OH)_2 + \Delta$$

Action of acids: Reacts with dilute mineral acids to form salt and water

Quicklime		Acid		Salt		Water
CaO	+	$2HCl$	→	$CaCl_2$	+	H_2O
CaO	+	$2HNO_3$	→	$Ca(NO_3)_2$	+	H_2O

Reaction with acidic oxides : Being basic oxide quicklime reacts with acidic oxides to form the respective salts.

Quicklime [basic oxide]		Acid oxide		Salt
CaO	+	SO_2	→	$CaSO_3$
$3CaO$	+	P_2O_5	→	$Ca_3(PO_4)_2$ [calcium phosphate]
CaO	+	SiO_2 [silica]	→	$CaSiO_3$ [calcium silicate]

Uses of quicklime

1. In manufacture of slaked lime
2. Quicklime is used in production of cement ,glass, sodalime(mixture of NaOH & CaO)
3. Quicklime acts as a drying agent for ammonia gas
4. In manufacture of limewater
5. In manufacture of milk of lime

Slaked lime[$Ca(OH)_2$]

Preparation

Method: By addition of water to quicklime (CaO)

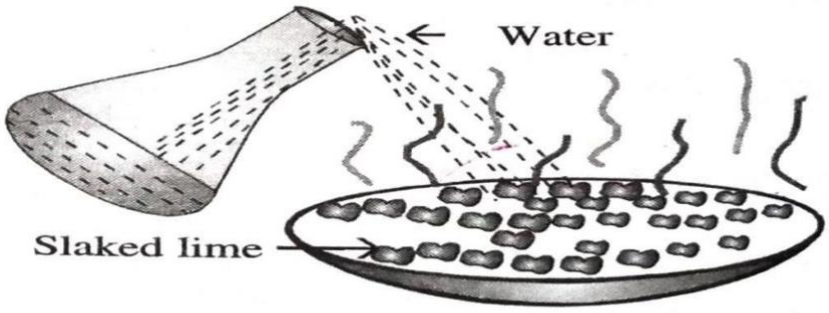
Equation $CaO + H_2O \longrightarrow Ca(OH)_2$(slaked lime) + $\triangle$

Procedure

When water is added to quicklime slowly the following observations are seen

- Nature of the reaction- Exothermic .heat evolved causes the waterboil with a hissing sound
- State of the substance: the quicklime swells
- State of the residue: the residue crumbles and turns into a white powdery mass with a clear watery solution forming at the top.[clear solution contains dissolved Ca(OH)2 and is called lime water]

Product the above solution is filtered and is dried to give a white mass of slaked lime



Preparation of slakedlime

Properties

Nature: **White amorphous powdery mass**

Stability: **Decomposes on strong heating to give calcium oxide and water which is slowly converted to steam and is expelled**

$$Ca(OH)_2 \longrightarrow CaO + H_2O$$

Solubility: **On addition of water slaked lime dissolves very sparingly .About 0.2% of slaked lime dissolves in water and the clear solution is called "Lime water"**

$$\underset{\text{[slaked lime] powder}}{Ca(OH)_2} + \underset{\text{[filter]}}{\text{water}} \rightarrow \underset{\text{[lime water] filtered solution}}{Ca(OH)_2 \text{ soln.}}$$

Action of acids: **Reacts with dilute mineral acids to form respective salts and water.**

Slaked lime	Acid	Salt	Water
$Ca(OH)_2$ +	$2HCl$ →	$CaCl_2$ +	$2H_2O$
$Ca(OH)_2$ +	$2HNO_3$ →	$Ca(NO_3)_2$+	$2H_2O$

Reaction with acidic oxides : Being a base calcium hydroxide reacts with ammonium salts liberating ammonia.

Slaked lime	*Ammonium salt*	*Salt + Water + Ammonia*
$Ca(OH)_2$ +	$2NH_4Cl$ →	$CaCl_2$ + $2H_2O + 2NH_3$
$Ca(OH)_2$ +	$(NH_4)_2SO_4$ →	$CaSO_4$ + $2H_2O + 2NH_3$
$3Ca(OH)_2$+	$2(NH_4)_3PO_4$→	$Ca_3(PO_4)_2$ +$6H_2O + 6NH_3$

***Note:** A suspension or a paste of slaked lime in water is called "milk of lime."*

Uses of quicklime

1. **slaked lime is used in the manufacture of bleaching powder[$CaOCl_2$],Caustic [NaOH],mortar**
2. **Salked lime is also used in softening temporary hard water [Clarke's process] ,dehairing hides from dead animals and in agriculture[neutralizes acidity of soil]**
3. **Lime water is used for detecting carbon dioxide ,skin lotions**
4. **Milk of lime is used for white washing buildings[carbon dioxide of the atmosphere reacts with milk of lime, forming a thin coating of calcium carbonate]**

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SELF EVALUATION

1. **Write down the reactants,reaction and purificaltion method involvedin the laboratory preparation of CO_2**
2. **Why is not dil H2SO4 used in preparation of CO_2?**
3. **Write the required conditions and the equation involved in the preparation of Urea from CO_2**
4. **What happens when a burning magnesium ribbon is held in a gas jar full of CO2.Write the equation**
5. **State the observation when CO_2 is passed through lime water.**
6. **What is dry ice?**
7. **What are the different types of fire extinguishers?**
8. **Differentiate between the two types of fire extinguisher based on their constituents and workings.**
9. **Which type of fire extinguisher should be used to extinguish electrical fires and why?**
10. **Define Carbon cycle.**

11. Differentiate between CO and CO_2 on the basis of i.vapour density,ii.Reaction with lime water, iii.Oxidising and reducing nature iv.combustibility

12. How can be CO separated from CO_2 by passing through conc. Potassium hydroxide solution?

13. Write the equation for preparation of quicklime.

14. What is limelight?

15. **Define slaking of lime with suitable equation.**

16. **Write the reaction between quicklime and dilute nitric acid.**

17. **What are the observation when water is slowly added to slaked lime?**

18. **Define limewater and milk of lime.**

19. **What happens when slaked lime is reacted with ammonium chloride?**

20. **how does slaked lime find usage in Clarke's process and in agriculture?**

END